

Muhammad Ali Waris Khan

AI Engineer | LLM & RAG | Python Backend | Data Engineering

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🌐 github.com/maw-khan 🔗 LinkedIn 🌐 Portfolio

Professional Summary

AI Engineer specializing in LLM applications, Retrieval-Augmented Generation (RAG), and Python backend development. Experienced in building AI systems with FastAPI, LangChain, Gemini AI, FAISS, and PostgreSQL, alongside hands-on experience in ETL, data engineering, web scraping, and Power BI analytics. Strong foundation in API development, document processing, data modeling, database systems, and production-oriented software engineering.

Education

Master's in Artificial Intelligence *NUST, Islamabad*
College of Electrical and Mechanical Engineering, National University of Sciences and Technology

BSc in Electrical Engineering *UET Taxila*
University of Engineering and Technology, Taxila

Professional Experience

Intern — NASTP (ExD) *Present*

- Worked hands-on with Python, PostgreSQL, database design, Power BI, data analysis, ETL workflows, and web scraping through practical technical assignments and projects.
- Developed Python-based ETL workflows involving data extraction, transformation, cleaning, validation, and loading into PostgreSQL using pandas and SQLAlchemy.
- Designed relational database structures and analytical workflows, followed by Power Query transformations, DAX measures, and interactive Power BI dashboards.
- Implemented web-scraping workflows using Requests and BeautifulSoup to retrieve, parse, extract, and export structured web data to CSV using pandas.
- Currently progressing through Generative AI and additional AI/data engineering tasks as part of the ongoing internship.

AI Intern — ACT AI National AI Training Programme *Jun 2026 – Jul 2026*

- Completed an intensive eight-week hands-on AI programme covering AI Foundations, Generative AI, Agentic AI, AI Tools & Productivity, and AI-enabled workflows.
- Applied AI concepts through practical assignments, assessments, and a final course project, developing experience with modern Generative AI tools and workflows.

AI/ML Engineering Intern — DevelopersHub Corporation *Apr 2026 – May 2026*

- Completed a six-week hands-on internship in AI/ML Engineering involving practical technical work and engineering-oriented learning.
- Demonstrated strong performance throughout the internship and contributed to assigned AI/ML engineering tasks and activities.

Lab Engineer (Academic) — NUST, Islamabad *Jan 2025 – Feb 2026*

- Assisted faculty members during laboratory sessions and practical coursework.
- Guided students in programming, debugging, and technical problem solving.
- Supported software setup, technical demonstrations, and computing resources for laboratory activities.

Skills

- **Technical Skills:** Python, SQL, Async Python, FastAPI, REST APIs, PostgreSQL, SQLAlchemy, ETL, RAG, LLM Applications, LangChain, LCEL, Gemini AI, FAISS, Embeddings, Semantic Search, Prompt Engineering, Data Modeling, Power BI, Power Query, DAX, Star Schema, Web Scraping
- **Tools:** pandas, NumPy, Beautiful Soup, Requests, lxml, PyMuPDF, Git, GitHub, Postman, Streamlit, VS Code
- **Professional Skills:** Problem Solving, Analytical Thinking, Debugging & Troubleshooting, Technical Communication, Research, Documentation, Collaboration

Certifications & Training

IBM Data Analyst Professional Certificate — Coursera

Aug 2026

11-course professional certificate covering Python, SQL, relational databases, data analysis, data visualization, Excel, Cognos Analytics, and practical data projects.

Selected Projects

1. AI Resume Analyzer with ATS Scoring

Python, FastAPI, Gemini AI, NLP, PyMuPDF

- Built an AI-powered resume analysis platform using LLM-based evaluation, keyword matching, and job-description comparison.
- Developed PDF extraction and preprocessing pipelines with validation and exception handling for resume and job-description documents.
- Created modular backend components and deployed the application using a FastAPI-based architecture.

2. YouTube Video RAG Chatbot

Python, FastAPI, LangChain, Gemini AI, FAISS

- Developed a Retrieval-Augmented Generation system for conversational querying of YouTube video transcripts.
- Implemented transcript extraction, document chunking, embeddings, vector search, contextual retrieval, and LLM-based answer generation.
- Built backend APIs for document processing and conversational interaction using a modular RAG architecture.

3. Python ETL & Data Pipeline — BookNook

Python, pandas, SQLAlchemy, PostgreSQL, ETL

- Built an end-to-end ETL pipeline to extract, transform, validate, and load customer, product, and order data into PostgreSQL.
- Used pandas and SQLAlchemy to clean datasets, map customer information, transform order data, and populate normalized relational tables.
- Designed database loading workflows for customers, products, orders, and order items, followed by analytical visualizations from the processed data.

4. MetroCare Healthcare Analytics Dashboard

PostgreSQL, SQL, Power BI, Power Query, DAX, Data Modeling

- Integrated and transformed multiple CSV datasets into a structured PostgreSQL database for healthcare analytics.
- Designed analytical data models and relational relationships to support hospital operations, patient, appointment, insurance, and revenue analysis.
- Built interactive Power BI dashboards using Power Query and DAX to analyze appointments, cancellations, insurance coverage, patient categories, and revenue-related KPIs.